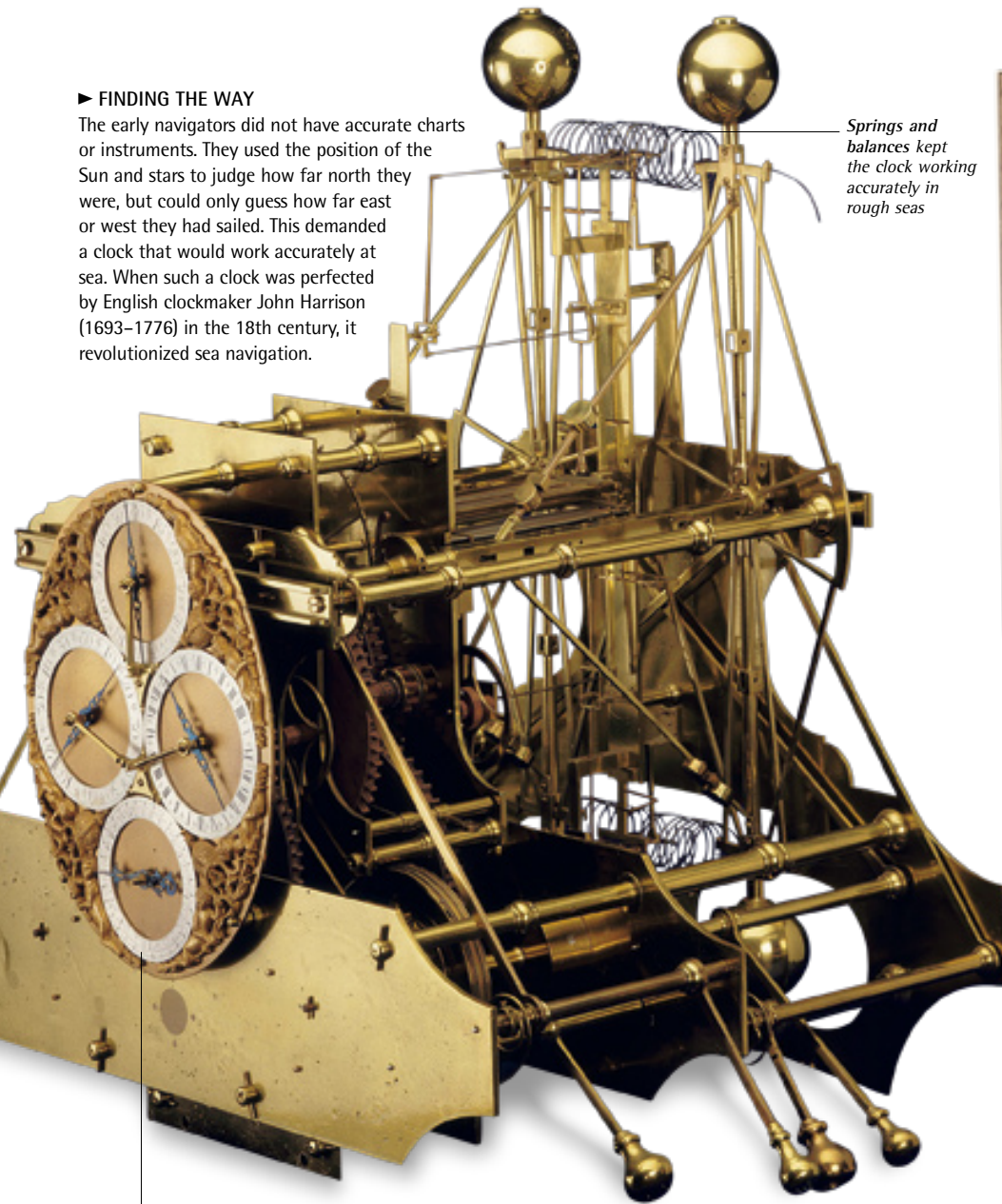


► FINDING THE WAY

The early navigators did not have accurate charts or instruments. They used the position of the Sun and stars to judge how far north they were, but could only guess how far east or west they had sailed. This demanded a clock that would work accurately at sea. When such a clock was perfected by English clockmaker John Harrison (1693–1776) in the 18th century, it revolutionized sea navigation.

Springs and balances kept the clock working accurately in rough seas



Dials for days, hours, minutes, and seconds were a feature of Harrison's first successful clock

► THE ENDS OF THE EARTH

The last of the world's oceans to be accurately surveyed were the icy waters of the Arctic and Antarctic. Many ships and crews were lost during attempts to find a northwest passage through the Arctic Ocean to Asia. The Southern Ocean around Antarctica was even more hazardous, but was gradually charted by explorers such as Charles Wilkes (1798–1877) and James Clark Ross (1800–1862). This painting shows Ross's two ships HMS *Erebus* and HMS *Terror* in 1842, threatened by towering Antarctic icebergs.



▲ CHARTING THE GLOBE

Accurate navigation demands accurate charts. Many of these were drawn up from data collected on long surveying expeditions in the 18th and 19th centuries. They included Captain James Cook's (1728–1779) explorations of the Pacific, during which he charted the eastern coast of Australia and many islands including New Zealand, shown here in Cook's own chart. Later survey voyages included those of HMS *Beagle* under Captain Robert FitzRoy (1805–1865) in the 1830s—a voyage later made famous by the observations of the ship's naturalist, Charles Darwin (1809–1882).